

Lightweight Crop Disease Recognition on Low-Cost Phones for Smallholder Farms

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Abstract—Smallholder farmers rarely have access to an agronomist when a crop disease appears. We present a compact convolutional network that recognises twelve rice and jute leaf diseases directly on a low-cost phone. The network is trained on 18,400 field photographs and reaches 93.1% accuracy while running in 41 ms per image without a network connection.

Index Terms—crop disease, mobile inference, convolutional networks, agriculture.

I. INTRODUCTION

Rice and jute are the two crops that most smallholder families in the delta depend on. Leaf diseases such as blast, brown spot and stem rot can remove a third of a harvest when they are noticed late. Extension officers are few, and a farmer may wait a week before somebody who can name the disease visits the field.

Phones with a camera are now common even in remote villages. This makes automatic recognition from a photograph attractive, but most published models are too large for the phones that farmers actually own, and they assume a fast connection to a server. Our goal is a model that fits in 6 MB and gives an answer on the device itself.

The contributions of this paper are: a field dataset collected over two growing seasons, a compact network designed for phones with 2 GB of memory, and a study with 64 farmers who used the application for one season.

II. RELATED WORK

Early systems classified leaf images taken under laboratory lighting [1]. Their accuracy falls sharply on field photographs, where leaves overlap and the background is soil or water [2]. Later work fine-tuned large image networks [3], which improved accuracy but produced models of several hundred megabytes. Pruning and quantisation reduce the size [4], yet few studies report timings on the inexpensive handsets common in rural markets.

III. METHODOLOGY

A. Data Collection

Photographs were taken by eleven trained volunteers in 37 villages between June 2021 and November 2022. Each photograph was labelled independently by two plant pathologists, and a third resolved disagreements. After removing blurred images the dataset holds 18,400 photographs in twelve disease classes and one healthy class.

B. Network Design

The network uses depthwise separable convolutions with a width multiplier of 0.5. We replace the final pooling layer with an attention block that weighs leaf regions more than background. Weights are quantised to eight bits after training. The final model occupies 5.7 MB.

TABLE I
COMPARISON OF MODELS ON THE FIELD TEST SET

Model	Size (MB)	Accuracy
ResNet-50	98	94.0
MobileNetV2	14	91.2
EfficientNet-B0	21	92.6
Ours	5.7	93.1

IV. EXPERIMENTS

We hold out the photographs from seven villages as a test set so that no field appears in both training and testing. All models are trained for 120 epochs with the same augmentation. Timings are measured on a phone with a 1.6 GHz processor and 2 GB of memory, averaged over 500 images.

Fig. 1. Confusion between brown spot and narrow brown spot accounts for most of the remaining errors.

V. RESULTS AND DISCUSSION

Table I compares our network with three common baselines. It is within one point of the largest model while being seventeen times smaller. On the phone it classifies an image in 41 ms, against 610 ms for ResNet-50. Farmers in the user study reported that an immediate answer in the field mattered more to them than the last point of accuracy.

Accuracy is lowest for the two brown spot diseases, which pathologists also find hard to separate from a photograph alone. When these two classes are merged, accuracy rises to 95.4%.

VI. LIMITATIONS

Our study has several limitations. First, all photographs come from a single river delta, so the model may not transfer to regions with different rice varieties or soil colour. Second, the dataset contains few images of early-stage infection, because farmers usually report a disease only after it has spread. Third, the user study lasted one growing season and involved 64 farmers, which is too small to measure any effect on yield.

Finally, we evaluated a single phone model. Devices with older cameras may produce images on which the network

performs worse than reported here.

VII. CONCLUSION AND FUTURE WORK

We presented a 5.7 MB network that recognises twelve leaf diseases on a low-cost phone with 93.1% accuracy and no network connection. A season-long study showed that farmers were able to use it without training.

In future work, we plan to collect photographs from two further regions and to add wheat and maize. We also intend to study whether early warnings change how much pesticide farmers apply.

DATA AVAILABILITY

The field photographs and labels that support the findings of this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.7654321>. The training code and the phone application are available at <https://github.com/riverside-agri/leaf-doctor-mobile> under the MIT licence.

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